

Supplemental Material for “Robust analog two-qubit gates from deep reinforcement learning: an independent reproduction and family-resolved assessment”

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This document collects material that supports the main text: the derivation of the leakage bound (Sec. S1), algorithmic details (Sec. S2), the complete configuration of every run (Sec. S3), the sweeps that fix the free parameters (Sec. S4), the horizon sweep of the direct optimizer (Sec. S5), the two family sweeps target by target (Sec. S6) and further remarks on the relation to hardware (Sec. S7). Section, equation, figure and table numbers without the prefix S refer to the main text.

S1. THE TSWT COHERENT LEAKAGE BOUND

We summarize the derivation of Eq. (5). Starting from the decomposition Eq. (3) in the regime $\epsilon/\Delta \ll 1$, one seeks an anti-Hermitian generator $\hat{S}(t) = \epsilon\hat{S}_1 + \epsilon^2\hat{S}_2 + \dots$ such that the transformed Hamiltonian Eq. (4) is block-diagonal to the desired order. Order by order, the block-off-diagonal components must satisfy

$$[\hat{H}_0, \hat{S}_1] = -\hat{H}_2, \quad [\hat{H}_0, \hat{S}_2] = -([\hat{H}_1, \hat{S}_1] - i\dot{\hat{S}}_1)_{\text{od}}, \quad (\text{S1})$$

which are solved elementwise on the off-diagonal blocks by dividing by the subspace energy differences $E_\alpha - E_{\alpha'}$; the assumption $\epsilon/\Delta \ll 1$ is what makes these divisions safe. After second order the residual block-off-diagonal Hamiltonian is

$$H_{\text{od}}^{(3)} = [\hat{H}_1, \hat{S}_2] + \frac{1}{3}[[\hat{H}_2, \hat{S}_1], \hat{S}_1] - i\dot{\hat{S}}_2 = O\left(\frac{\epsilon^3}{\Delta^2}\right). \quad (\text{S2})$$

The leakage amplitude out of Ω_0 over $[0, T]$ is then an interaction-picture integral of $\hat{H}_{\text{od}}$ against a rapidly oscillating phase $e^{i \int \Delta dt}$. Repeated integration by parts converts this into boundary terms, which pick up $\|\hat{H}_{\text{od}}\|/\Delta$ at $t = 0$ and $t = T$, plus a residual integral controlled by the second derivative of $\hat{H}_{\text{od}}$ divided by Δ^2 . Collecting these gives Eq. (5).

The two contributions dominate in complementary regimes: for slow, off-resonant modulation the boundary terms dominate, and for faster on-resonant modulation the derivative term does. This is why the bound is usable as a single penalty across the operating range rather than only in one limit.

S2. ALGORITHMIC DETAILS

Environment step. Given the current observation and propagator: map the action to physical controls; apply the bandwidth filter Eq. (12); if stochastic training is enabled, sample Gaussian perturbations for $g, \delta_{1,2}, f_{1,2}$ and η and wrap the phases modulo 2π ; build H_t from Eq. (1); propagate $U_{t+1} = e^{-iH_t\Delta t}U_t$ by exact diagonalization; update the leakage estimator with the same, possibly noisy, H_t ; evaluate the running cost Eq. (7) and reward $r = -C$; test the stopping condition (time limit or cost threshold); emit the next observation Eq. (13).

Incremental leakage estimator. At each step: decompose H_t into H_0, H_1, H_2 ; solve Eq. (S1) for S_1 ; estimate $\dot{S}_1$ from the stored previous value by finite differences; solve for S_2 ; estimate $\dot{S}_2$; form $H_{\text{od}}^{(3)}$ from Eq. (S2) and symmetrize; take its spectral norm. The first and last samples update the boundary contribution; the discrete second derivative, formed from the two previous stored values, updates the integral contribution. Storing only the two previous states keeps the estimator $O(1)$ per step.

TRPO loop. Per iteration: collect a batch of stochastic rollouts in parallel workers; compute discounted returns and generalized advantage estimates [1]; take the KL-constrained policy step Eq. (9) by conjugate gradient with backtracking line search; refit the value network by regression on the empirical returns; evaluate the deterministic policy in the nominal environment and store the pulse if it improves on the best so far. Checkpoints and induced pulses are saved every iteration, so that comparisons can be made at the level of the realized pulse rather than only at the level of network weights.

Direct optimizer. For each candidate horizon the raw parameters are initialized to zero, which after the tanh map gives zero amplitudes and phases π . Each of the 400 steps filters the controls, simulates one trajectory perturbed at σ_{train} , evaluates the UFO cost including the leakage bound on that noisy trajectory, back-propagates, clips the gradient norm at 10 and takes an Adam step. Within a horizon the stored pulse is the iterate with the lowest sampled loss; Secs. III C and S5 describe how the horizon is selected.

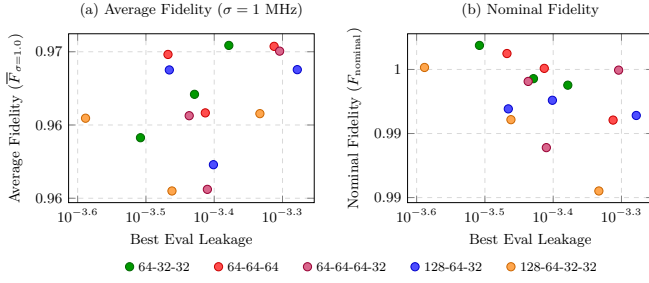


FIG. S1. Architecture trade-offs: evaluation leakage against (a) average fidelity at $\sigma_{\text{noise}} = 1$ MHz and (b) nominal fidelity. Leakage minimization is not a proxy for robustness — the layout with the lowest leakage, (128, 64, 32, 32), has the worst aggregate cost and the weakest robustness.

S3. COMPLETE CONFIGURATION

Table S1 collects every setting used. We report per-experiment values rather than a single idealized configuration, because the runs differ in budget and stopping cost, and the two single-target agents were trained in two stages.

S4. FIXING THE FREE PARAMETERS

The two sweeps summarized in Sec. IIIE were run on the representative target $N(2.2, 2.2, \pi/2)$, under a reduced budget and before any production run. Every sweep run is noise-trained at $\sigma_{\text{train}} = 1$ MHz.

1. Actor-critic architecture

We swept five hidden-layer layouts of comparable scale, three seeds each, at 20 policy iterations and 5000 episodes per batch (Table S2). Selection was not made on total cost alone, because a network can reach a favourable cost by trading leakage against fidelity or by settling on an unnecessarily long pulse; we therefore tracked nominal fidelity, leakage, runtime and robustness at $\sigma_{\text{noise}} = 1$ MHz simultaneously.

Two conclusions follow. First, architecture is not neutral: moderate changes in width and depth produce measurable differences in the final trade-off. Second, the dependence is not monotonic. The deepest model tested achieves the *lowest* mean leakage of the sweep and simultaneously the *worst* aggregate cost, the weakest robustness and the lowest nominal fidelity (Fig. S1). Minimizing one term of a multi-objective cost is not a selection principle.

The best aggregate performer is (64, 64, 64), but its margin over the paper layout (64, 32, 32) is 0.2% in evaluation cost, well inside the seed-to-seed spread: the three seeds of (64, 32, 32) alone reach $\bar{F}_{\sigma=1} = 0.959, 0.962$ and 0.965. We therefore retain the paper architecture

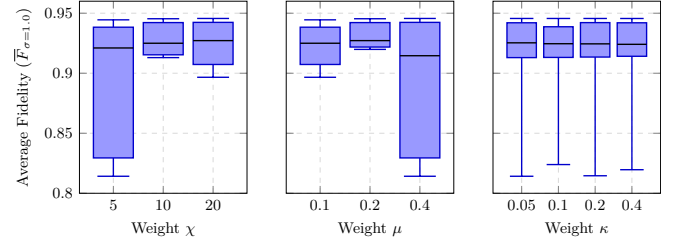


FIG. S2. Sensitivity of the average fidelity at $\sigma_{\text{noise}} = 1$ MHz to the UFO weights, shown separately for the fidelity weight χ , the boundary weight μ and the runtime weight κ . Underweighting fidelity ($\chi = 5$) opens a failure mode near $\bar{F} \approx 0.82$; $\mu = 0.4$ is strongly bimodal; κ has almost no effect in this fixed-horizon regime.

(64, 32, 32) for all production runs, which maximizes comparability with Ref. [2]. The same spread sets the resolution of every single-seed comparison in the paper: differences of a few 10^{-3} in $\bar{F}$ at 1 MHz between two individual runs are not significant on their own.

2. Sensitivity to the UFO weights

The weights in Eq. (7) are part of the physical definition of the problem, not optimizer settings, so they deserve separate treatment. We swept $\chi \in \{5, 10, 20\}$, $\mu \in \{0.1, 0.2, 0.4\}$, $\kappa \in \{0.05, 0.1, 0.2, 0.4\}$ at fixed $\beta = 10$ (36 configurations, two seeds each, five iterations of 10 000 episodes with stopping cost 0.15). Holding β fixed makes the sweep a study of the ratios χ/β , μ/β , κ/β , which is the physically meaningful reading and keeps the design tractable.

A methodological caveat governs the whole sweep: the best evaluation cost is *not* a valid selection criterion across configurations, because different weights define different scalarizations and their numerical costs are not commensurable. Since all runs here terminate at the same evaluation time of 180 ns, increasing κ mostly adds a rigid offset. We therefore compare on physical quantities only — nominal fidelity, leakage, and robustness at $\sigma_{\text{noise}} = 1$ MHz.

The fidelity weight has the clearest effect (Fig. S2). At $\chi = 5$ the distribution is broad and several runs collapse to $\bar{F} \approx 0.81$ –0.83; at $\chi = 10$ and $\chi = 20$ it tightens around 0.92–0.95. Underweighting logical accuracy relative to leakage lets the optimizer settle for controls that are clean but simply not the right gate. The boundary weight behaves non-monotonically: $\mu = 0.2$ gives the best median trade-off, $\mu = 0.1$ leaves slightly more leakage, and $\mu = 0.4$ becomes strongly bimodal, producing both some of the best runs of the sweep and its worst lower-tail failures. Larger μ does reduce the leakage proxy on average — the lowest-leakage region is often at $\mu = 0.4$ — but this does not translate into better robust control, which is the same lesson the architecture sweep taught. The runtime weight κ is nearly inert here,

TABLE S1. Complete configuration, as actually run. The four right-hand columns are the two single-target RL agents (Secs. IV–V), the architecture and UFO-weight sweeps (Sec. S4) and the curriculum runtime sweep (Sec. VI); an entry spanning several columns applies to all of them. Every RL run except the noise-free agent was trained with noise.

Quantity	Symbol	Single target	Arch. sweep	Weight sweep	Curriculum
<i>Physical model</i>					
Levels per mode	—			3	
Anharmonicity	η			−200 MHz	
Control bound	$u_{\max}$			20 MHz	
Filter bandwidth	B			10 MHz	
Timestep	Δt			2 ns	
Max gate duration	$T_{\max}$			180 ns	
Runtime norm.	T_{norm}			60 ns	
Target	—		$N(2.2, 2.2, \pi/2)$		$\alpha = 0, 0.157, \dots, 3.14$
<i>UFO weights</i>					
Fidelity, leakage	χ, β	10, 10	10, 10	{5, 10, 20}, 10	10, 10
Boundary	μ	0.2	0.2	{0.1, 0.2, 0.4}	0.2
Runtime	κ	0.1	0.1	{0.05, 0.1, 0.2, 0.4}	0.1
<i>TRPO</i>					
Hidden sizes	—	(64, 32, 32)	five layouts	(64, 32, 32)	(64, 32, 32)
Trust region	δ_{KL}			0.01	
Discount, GAE parameter	$\gamma_{\text{RL}}, \lambda$			0.99, 0.97	
CG iterations, damping	—			10, 0.1	
Value lr, epochs	—			10^{-3} , 50	
Initial exploration	$\log \sigma_0$	−0.5	−1.0	−1.0	−0.5
Episodes per batch	—	10 000 ^a	5000	10 000	4000
Policy iterations	—	100 + 15 / 100 + 5 ^a	20	5	≤150 per target
Stopping cost	C_{term}	0.40, then 0.30 ^a	0.40	0.15	0.40 ^b
Training noise	σ_{train}	0 / 1 MHz ^c	1 MHz	1 MHz	1 MHz
Seeds	—	1	1, 2, 3	1, 2	2
<i>Direct optimizer (single target and family sweep)</i>					
Iterations per horizon, lr	—		400, 0.03; one deterministic initialization; seed 2		
Horizon grid	—		40–180 ns, step 10 ns		
Training noise	σ_{train}		1 MHz, one noisy trajectory per step		
Family targets	—		$\alpha = 0, 0.1, \dots, 3.1$ and π		
<i>Robustness evaluation</i>					
Dense grid	σ_{noise}	0.1–3.5 MHz, step 0.001 MHz, 60 samples; EWMA spans 50 / 80			
Matched trajectories	σ_{noise}	11 values in 0.1–3.5 MHz, 200 samples each			
Closed-loop test	σ_{noise}	1 MHz, 100 samples			

^a Both agents: iterations 1–100 with 10 000 episodes per batch and $C_{\text{term}} = 0.40$, from the same seed. They then continued with $C_{\text{term}} = 0.30$: the noise-free agent for 15 iterations with 5000 episodes per batch, the noise-trained one for 5 iterations with 10 000. The three further pairs of Sec. VD (seeds 1, 2, 3) have the settings of iterations 1–100, with the leakage bound of the noise-trained agents accumulated from the noise-free controls.

^b Also the advancement threshold. The cold-start target $\alpha = 0$ used a hand-tuned sequence of stopping costs.

^c Noise-free / noise-trained agent.

TABLE S2. Architecture sweep on $N(2.2, 2.2, \pi/2)$, averaged over three seeds. Best value in each column in bold.

Hidden sizes	$\overline{C}_{\text{eval}}$	$\overline{F}_{\text{eval}}$	$\overline{F}_{\sigma=1}$	$\overline{T}$ (ns)	$\overline{L}$ (10^{-4})
(64, 64, 64)	0.3584	0.9941	0.9637	106.7	4.05
(64, 32, 32)	0.3592	0.9950	0.9622	111.3	3.67
(64, 64, 64, 32)	0.3626	0.9926	0.9604	107.3	4.17
(128, 64, 32)	0.3630	0.9920	0.9616	106.7	4.22
(128, 64, 32, 32)	0.3691	0.9906	0.9589	110.7	3.56

for the structural reason noted above; its influence reappears in Sec. VI, where gate duration is the observable rather than a fixed boundary condition.

The Pareto-favourable region is occupied jointly by $(\chi, \mu) \approx (10, 0.2)$ and $(\chi, \mu) \approx (20, 0.4)$, with the latter marginally ahead. Given two seeds and a small budget, that margin does not justify departing from the refer-

ence scalarization, so we retain the weights of Eq. (16) throughout.

S5. HORIZON SWEEP OF THE DIRECT OPTIMIZER

The direct optimizer is trained under noise, but its pulses are compared in the deterministic simulator. Under that re-evaluation the best horizon is $T^* = 70$ ns, with nominal cost 0.1576, fidelity 0.99662 and leakage 2.82×10^{-4} . The optimizer’s own noisy objective selects 60 ns instead, a pulse with cost 0.3806, fidelity 0.98205 and leakage 3.67×10^{-4} ; Appendix B explains why it prefers the shorter horizon.

The horizon dependence (Fig. S3) shows what sets the optimum. Below 60 ns the pulses fail, with fidelity 0.73

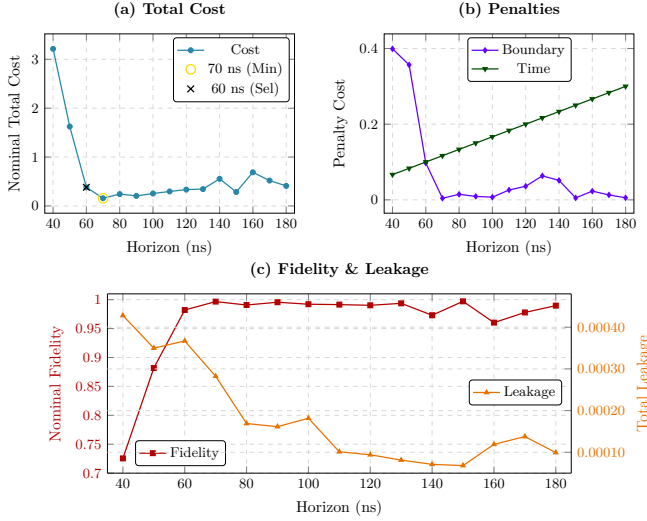


FIG. S3. Direct optimizer on $N(2.2, 2.2, \pi/2)$ versus gate horizon. (a) Nominal total cost, with the optimum at 70 ns and the pulse selected by the noisy objective at 60 ns both marked. (b) Boundary and runtime penalties. (c) Nominal fidelity and total leakage. The optimum arises from the simultaneous collapse of the boundary term and the rise in fidelity; longer horizons saturate in fidelity and are penalized by runtime.

at 40 ns and 0.88 at 50 ns, although their leakage stays in the 10^{-4} range: under strong time compression a pulse cannot be both accurate and zero at its ends, and it is the boundary term, not the leakage, that grows. Between 60 and 70 ns that term collapses and the cost falls from 0.381 to 0.158. Longer pulses are slightly more accurate and less leaky (fidelity 0.9971 at 150 ns) but pay for it in runtime, so the optimum is the best-balanced pulse and not the most accurate one. The 70 ns pulse is 3.1 times faster than the generic compiled reference of Eq. (11) and 2.1 times faster than the two-CNOT synthesis of Eq. (20).

S6. THE TWO FAMILY SWEEPS, TARGET BY TARGET

Table S3 lists the curriculum sweep of Sec. VI, and Fig. S4 shows the independent sweep of the direct optimizer.

S7. RELATION TO HARDWARE

This paper compares optimizers on a common model, and none of its fidelities is a statement about a device. Five remarks place that model against current hardware.

Noise. Frequency and amplitude noise in superconducting circuits is dominated by low-frequency components — $1/f$ flux and charge noise and slow drift of the calibration [3, 4]. Of the noise models of Sec. V the quasi-static one is therefore the closer to a device, and the white

TABLE S3. Curriculum runtime sweep over $N(\alpha, \alpha, \pi/2)$. $\varepsilon = \alpha - \pi/2$ is the exchange angle of Eq. (19) and T the final evaluation runtime; the last two columns give the speedup over the generic reference, 215 ns, and over the target-specific synthesis time of Eq. (20), 150 ns for every row shown. Rows marked † did not converge (final cost above 0.40): their roll-outs run to the 180 ns ceiling and no speedup is quoted. The row marked ‡ is the identity target.

α	ε	T (ns)	F^{nom}	$\bar{F}_{\sigma=1}$	215/ T	$T_{\text{syn}}^{\text{tgt}}/T$
0.000†	-1.571	180	0.5577	0.6133	—	—
0.157	-1.414	142	0.9880	0.9510	1.51	1.06
0.314	-1.257	140	0.9905	0.9533	1.54	1.07
0.471†	-1.100	180	0.9538	0.9179	—	—
0.628†	-0.943	180	0.9480	0.9149	—	—
0.785†	-0.786	180	0.9492	0.9055	—	—
0.942	-0.629	148	0.9905	0.9504	1.45	1.01
1.099	-0.472	76	0.9850	0.9674	2.83	1.97
1.256	-0.315	140	0.9894	0.9535	1.54	1.07
1.413	-0.158	132	0.9880	0.9542	1.63	1.14
1.570‡	-0.001	2	0.99999	0.9994	—	—
1.727	0.156	124	0.9861	0.9575	1.73	1.21
1.884	0.313	108	0.9821	0.9582	1.99	1.39
2.041	0.470	106	0.9825	0.9595	2.03	1.42
2.198	0.627	110	0.9837	0.9545	1.95	1.36
2.355	0.784	118	0.9863	0.9573	1.82	1.27
2.512	0.941	136	0.9919	0.9567	1.58	1.10
2.669	1.098	144	0.9918	0.9547	1.49	1.04
2.826†	1.255	180	0.9340	0.8944	—	—
2.983†	1.412	180	0.9213	0.8872	—	—
3.140†	1.569	180	0.9400	0.9020	—	—

model of Ref. [2], under which no training can help, the further.

Decoherence. The model is closed. Weak relaxation and pure dephasing, at rates $\Gamma_1 = 1/T_1$ and $\Gamma_\varphi = 1/T_\varphi$ on each qubit, lower the average fidelity of any two-qubit gate of duration T by $\frac{4}{5}T(\Gamma_1 + \Gamma_\varphi)$, whatever the gate [5]: like white control noise, Markovian decoherence leaves a pulse nothing to exploit but its duration. For $T_1 = T_\varphi = 100 \mu\text{s}$, of the order of current processors [6], this is 1.1×10^{-3} for the 70 ns pulse of the direct optimizer and 1.9×10^{-3} for the 116 ns pulse of the noise-free agent, comparable to the nominal infidelities of Table I. By the same formula the white noise of the model at 1 MHz costs as much as $T_1 = T_\varphi \approx 5 \mu\text{s}$, and the runtime weight of Eq. (7) corresponds to $T_1 = T_\varphi \approx 12 \mu\text{s}$. Both describe devices an order of magnitude less coherent than today's; a weight matched to current coherence would move the optimum of Eq. (7) towards longer and more accurate pulses.

Bandwidth. The 10 MHz filter is narrow by hardware standards. Sycamore processors, descendants of the gmon, drive single-qubit gates with 25 ns pulses and perform iSWAP-like gates by turning on a 20 MHz coupling for 12 ns [7]: the amplitude bound of the present model, four to six times faster than the 54–72 ns that its filter allows for the same exchange angle (Sec. VI E). The run-times of Sec. VI are thus set by a modelling choice of Ref. [2] more than by the coupler.

Higher levels. Those fast gates are not members of

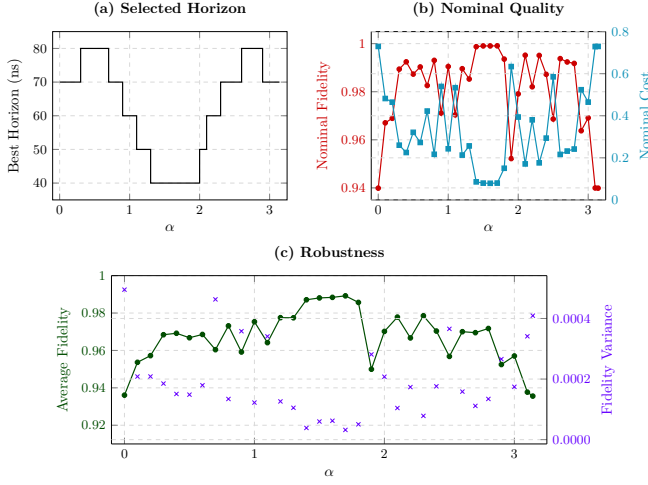


FIG. S4. No-transfer control: the direct optimizer swept independently over 33 targets of $N(\alpha, \alpha, \pi/2)$. (a) Selected horizon versus α . (b) Nominal fidelity and cost. (c) Average fidelity at $\sigma_{\text{noise}} = 1$ MHz and the corresponding variance. The horizon is shortest around the identity target $\alpha = \pi/2$ and grows with the exchange angle $|\alpha - \pi/2|$.

the benchmark family. The coupling shifts $|11\rangle$ through $|02\rangle$ and $|20\rangle$ by $4g^2/|\eta|$, so an exchange pulse carries a conditional phase $\phi = (4/|\eta|) \int g^2 dt$, which $N(\alpha, \alpha, \pi/2)$ does not contain and local rotations cannot remove; an otherwise perfect pulse is left with $1 - F = \sin^2(\phi/4)$. This is what limits the pulses optimized without noise in Sec. VF, whose nominal infidelities equal $\sin^2(\phi/4)$ to within 1%. In this model the fidelity of an exchange pulse therefore improves only as $1/T^2$, a trade-off that native gate sets avoid by making ϕ part of the gate [8] or by cancelling it through the level structure of the coupler [9].

Absolute fidelity. Two-qubit gates on tunable-coupler transmons reach a Pauli error of 3.8×10^{-3} for a continuous fSim family on gmon qubits [8], and fidelities of 99.76% and 99.87% for CZ and iSWAP gates [9], decoherence included. The closed-system fidelities of Table I are not competitive pulse designs and are not offered as such: what transfers from this study is the comparison between methods and the two structural results of Sec. VIII.

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